

Rare-Event Recurrence Diagnostics for Econometric Stress Analysis

Author Name
Institution
`author@example.com`

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Abstract

This article presents a reproducible computational framework for studying rare-event recurrence in macro-financial time series. The contribution is deliberately descriptive rather than causal: the repository implements threshold exceedance extraction, runs-based extremal-index summaries, return-time diagnostics, multivariate stress summaries, and comparisons with standard volatility-style baselines. The strongest current support comes from synthetic and benchmark workflows, which show how processes with similar large observations can differ in clustering and recurrence structure. The empirical workflow is presented as an application template with controlled data-loading, catalog, and cache-refresh paths, rather than as a source of standalone macroeconomic conclusions. The resulting claim is modest but useful: recurrence-oriented rare-event diagnostics provide a disciplined way to distinguish isolated extremes from repeated returns to dangerous regions.

Keywords: rare events, recurrence diagnostics, extremal index, return times, macro-financial stress

1 Contribution Summary

The repository currently supports three article-level contributions:

1. a reproducible synthetic comparison showing that large events can differ in recurrence and clustering structure even when they look superficially similar;
2. a compact diagnostic workflow for exceedances, return times, extremal-index summaries, and multivariate stress timing;
3. an empirical application template with explicit data-loading, catalog, and cache-refresh paths that is ready for disciplined use on real public datasets.

It does not currently support causal attribution, structural macroeconomic estimation, deterministic-law claims, or exact crisis-date prediction. Those boundaries should remain explicit throughout the article.

2 Introduction

Econometric analyses of stress usually center on shocks, volatility, breaks, and regime changes. Those are essential objects. The narrower question pursued here is different: when an extreme event occurs, is it better described as an isolated exceedance or as part of a recurring pattern of returns to a dangerous region?

That distinction is central because social and financial damage is often driven less by one large observation than by recurrence and clustering. A single inflation surprise, one quarter of severe stress, or one volatility burst may be absorbed. Repeated returns to the same dangerous region are harder to absorb. They affect expectations, financing conditions, policy credibility, and the distribution of future risk.

This repository is built around that narrower question. It uses rare-event tools associated with dynamical-systems extreme value theory to study exceedances, return times, recurrence, clustering, and joint stress. The claim is deliberately modest. The repository does not claim that observed economies follow literal deterministic laws. It claims only that recurrence structure in rare events can be measured, summarized, and compared in a reproducible way.

3 Why Recurrence Deserves Separate Attention

Standard econometrics remains the baseline. Autoregressive models summarize persistence. Volatility models describe changing second moments. Break models and regime methods detect structural change. Those tools are useful and remain necessary.

The limitation is not that they are wrong. The limitation is that crisis-relevant tail behavior is often reduced to large observations inside a broader stochastic model, while the pattern of repeated return to a stress region receives less direct attention than it deserves.

For risk management and policy interpretation, that pattern matters. Two processes may have similar marginal tail behavior and still differ sharply in how extremes cluster through time. If one process produces isolated exceedances and another produces repeated tail occupation, the practical meaning of danger is different even when conventional summaries look similar.

4 Conceptual Translation

The dynamical-systems language is useful because it translates cleanly into applied diagnostic work.

Dynamical-systems idea	Econometric interpretation
observable along an evolving path	measured economic or financial series
rare set	crisis or stress region
hitting time	time until a stress region is reached
return time	time between repeated visits
clustering	persistence of extreme episodes
extremal index	compact summary of whether extremes arrive in clusters
joint recurrence	multivariate stress across variables

Table 1: Conceptual translation from recurrence language to econometric diagnostics.

This translation does not require a strong structural claim about the economy. It only requires a willingness to ask whether observed tail events recur in a patterned way that is worth measuring directly.

5 What the Repository Actually Implements

The current codebase supports a reproducible rare-event workflow with the following components:

- threshold exceedance extraction;
- runs-based extremal-index summaries;
- return-time and recurrence diagnostics;
- multivariate joint-stress summaries;
- descriptive comparison with baseline volatility-style diagnostics;
- saved article-facing figures with matching source CSV files.

These are the supported outputs. They are sufficient for a computational article about recurrence diagnostics. They are not sufficient for stronger claims such as causal identification, structural macroeconomic estimation, or exact crisis-date prediction.

The stable article-facing assets are:

- `01_conceptual_pipeline.png` for the workflow from raw inputs to article evidence;
- `02_synthetic_extremes_comparison.png` for clustered versus more isolated synthetic extremes;
- `03_return_time_distribution.png` for descriptive return-time behavior;
- `04_extremal_index_by_series.png` for cross-series clustering comparison;

- `05_multivariate_stress_timeline.png` for joint-stress timing;
- `06_econometric_vs_recurrence_diagnostics.png` for descriptive comparison between baseline volatility-style summaries and recurrence-oriented diagnostics.

Each of these figures has a matching source CSV, which means the article can point to saved structured outputs rather than to ad hoc plotting state.

6 Computational Workflow and Reproducibility

The repository is designed as a package-first computational workflow rather than as a loose notebook collection. That matters for the article because the evidentiary unit is not only a plotted figure but a chain of reproducible objects: configuration, loader choice, saved tables, saved figures, and matching source CSV assets.

At a high level, the workflow is:

1. load or simulate a panel in the repository’s standard long format;
2. define a threshold-based stress region;
3. extract exceedances and recurrence summaries;
4. compute clustering diagnostics such as the runs-based extremal index;
5. save article-facing outputs with matching CSV sources.

The article therefore rests on saved computational artifacts rather than on transient notebook state. That design is especially important for recurrence diagnostics, where small implementation differences in thresholding or declustering can materially alter the interpretation.

Figure asset not bundled in this working tree.
Expected file: `figures/01_conceptual_pipeline.png`

Figure 1: Conceptual pipeline from input panel to article-facing evidence. This figure summarizes the computational workflow used in the repository: data loading or simulation, threshold definition, exceedance extraction, recurrence and clustering diagnostics, and saved article-facing outputs. It illustrates the analysis structure rather than an empirical result.

7 Synthetic Evidence as the Current Strongest Support

The strongest fully supported evidence in the repository is synthetic. That is not a weakness if it is stated honestly. Synthetic systems are the cleanest way to show why recurrence structure matters, because they let us compare processes that differ in clustering behavior without pretending that the comparison is already an empirical finding.

The repository includes controlled generators for familiar stochastic baselines and nonlinear benchmark systems. Those workflows support a simple but important point: similar-looking series can differ materially in recurrence and clustering, and that difference appears in exceedance patterns, return times, and extremal-index summaries.

In article terms, that synthetic support is most directly expressed through `02_synthetic_extremes_comparison.p` which is the clearest visual anchor for the claim that tail behavior differs not only in magnitude but also in recurrence structure.

Figure 2 should therefore be read as the article’s main methodological evidence. It is not offered as a model of an actual economy. It is offered as a controlled comparison showing why recurrence-sensitive diagnostics belong in the econometric discussion at all.

This is exactly the right use of the synthetic layer. It is not evidence that economies are chaotic maps. It is evidence that recurrence diagnostics detect behavior not fully summarized by means, variances, or one-dimensional tail thickness alone.

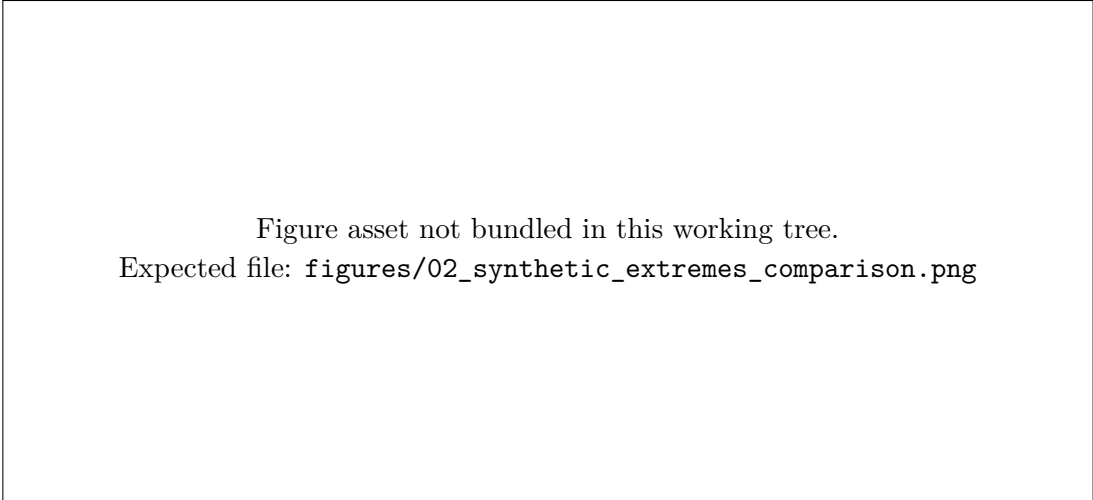


Figure asset not bundled in this working tree.
Expected file: `figures/02_synthetic_extremes_comparison.png`

Figure 2: Synthetic comparison of clustered and more isolated extremes. This figure compares benchmark processes with different recurrence structure. It supports the methodological claim that similar-looking large observations can differ materially in clustering behavior. It does not constitute empirical evidence about a specific economy.

8 Extremal Index as a Practical Diagnostic

Among the implemented summaries, the extremal index is especially useful. Informally, it distinguishes between extremes that behave more like isolated episodes and extremes that arrive in clusters. That distinction is often more decision-relevant than another small refinement in average behavior.

In a macro-financial context, clustered exceedances matter because they correspond more closely to sustained stress than to isolated bad draws. A regulator, treasury, or central bank may care less about one excursion beyond a threshold than about repeated short-horizon returns to the same stress region.

The value of the extremal index in this repository is therefore operational rather than metaphysical. It is a compact description of whether the tail is episodic or persistent.

The most direct article-facing support for this point is `04_extremal_index_by_series.png`, which summarizes cross-series clustering differences without requiring a stronger structural interpretation than the code currently supports.

In article form, the safest reading of Figure 4 is comparative rather than absolute. It highlights differences in clustering behavior across series or benchmark observables; it does not establish a universal economic threshold for crisis persistence.

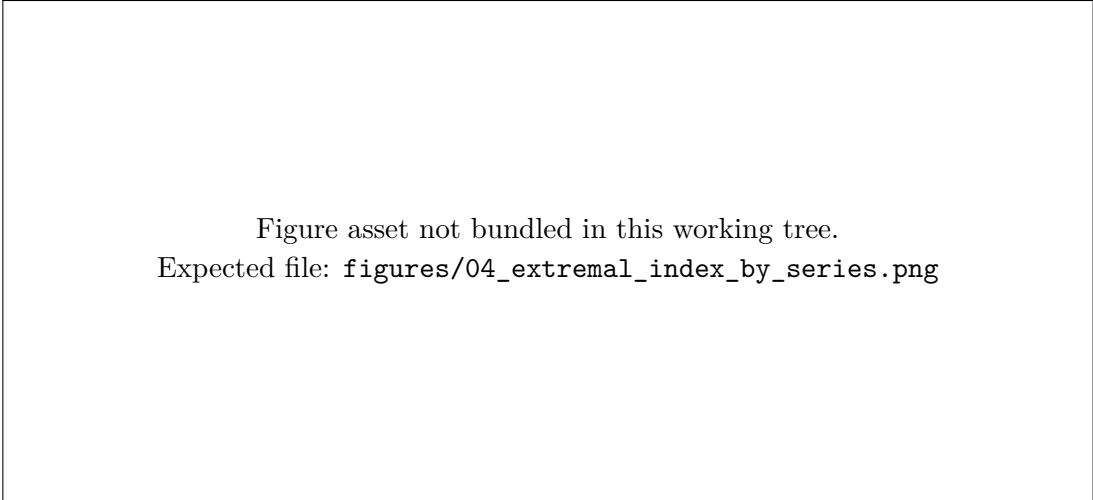


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Expected file: `figures/04_extremal_index_by_series.png`

Figure 3: Extremal-index comparison across series or benchmark observables. This figure reports cross-series differences in clustering diagnostics. Lower values indicate more clustered extreme episodes in the sample or benchmark workflow. The figure supports comparative interpretation, not universal threshold rules.

9 Return Times and Recurrence Summaries

Return-time summaries complement clustering diagnostics. They ask how long the process typically takes to revisit a threshold-defined stress region. Again, the claim must remain careful: these summaries describe recurrence patterns in observed or simulated data; they do not deliver exact forecasting rules for the next crisis date.

Used correctly, they help distinguish between environments where stress regions are rarely revisited and environments where the system returns quickly and repeatedly. That is a meaningful descriptive difference even when it does not identify the cause of the behavior.

That descriptive role is exactly what `03_return_time_distribution.png` supports. It is evidence about recurrence timing patterns, not a device for exact crisis-date prediction.

The article should therefore speak of return-time summaries as timing diagnostics, not as forecasting devices. They help organize how stress reappears in the sample; they do not tell the reader when the next crisis will occur.

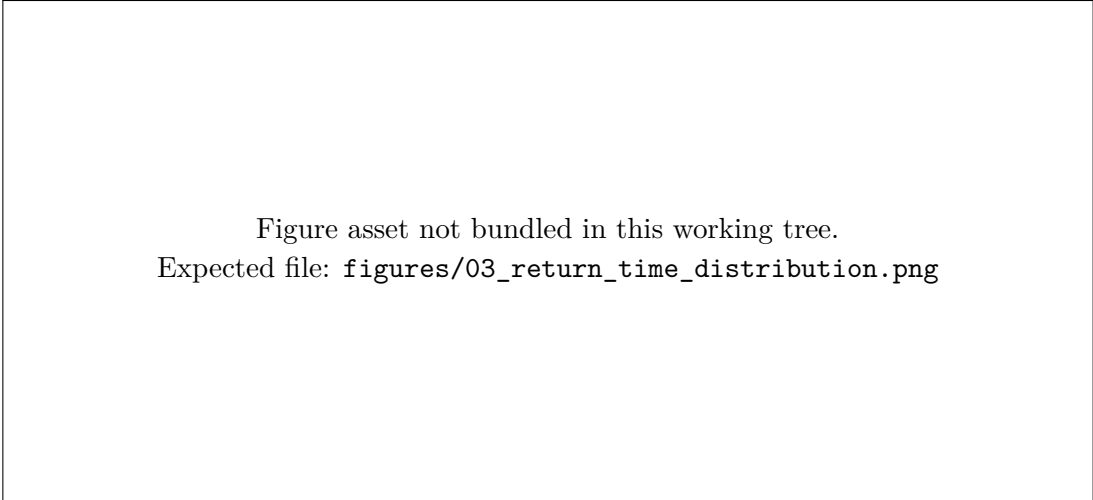


Figure asset not bundled in this working tree.
Expected file: `figures/03_return_time_distribution.png`

Figure 4: Return-time distribution for a benchmark observable. This figure summarizes descriptive return-time behavior for repeated visits to a threshold-defined stress region. It should be read as a recurrence diagnostic, not as a forecast of the next crisis date.

10 Multivariate Stress and Joint Recurrence

The repository also supports multivariate stress summaries. This matters because macro-financial danger often appears through joint pressure rather than through one variable in isolation. A single stressed series may be manageable; simultaneous stress across inflation, spreads, funding proxies, or market indicators is more consequential.

The supported interpretation here is still descriptive. Joint recurrence diagnostics can show that stress appears together across observables more often than an isolated reading would suggest. They do not, by themselves, identify a structural transmission mechanism.

The relevant stable visual here is `05_multivariate_stress_timeline.png`, which supports a joint-stress timing narrative but not a causal account of why the co-movement occurs.

This is the correct place in the article to emphasize that multivariate stress can be descriptively important even when the transmission channel remains unidentified. Co-occurrence is not causation, but it is still analytically relevant.

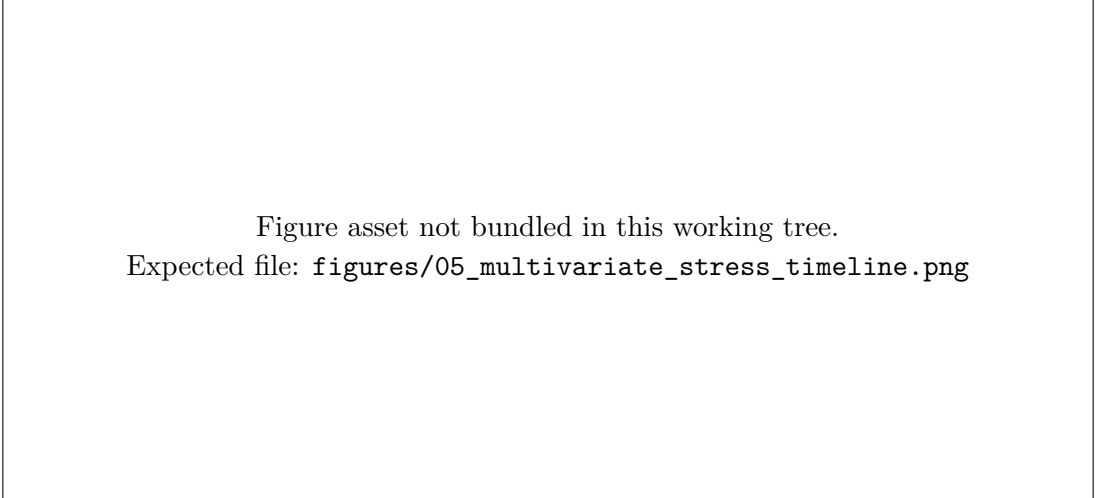


Figure asset not bundled in this working tree.
Expected file: `figures/05_multivariate_stress_timeline.png`

Figure 5: Multivariate stress timeline. This figure shows the timing of joint stress across observables in a multivariate setting. It supports descriptive claims about co-occurrence and joint recurrence, but it does not identify a causal transmission mechanism.

11 Data and Empirical Application Template

The repository now supports several empirical input modes: direct local files, per-series configuration blocks, catalog-based materialization, multi-catalog workflows, and cached remote-source patterns. That makes the empirical side of the project operationally serious enough to document, but not yet strong enough to treat as finished empirical evidence in the absence of a specific validated dataset.

For that reason, the article should present the empirical workflow as a template for disciplined application rather than as a completed case study. The safe statement is that the code can now take a real public dataset through a controlled loading and recurrence-analysis pipeline. The unsafe statement would be that the repository already establishes a substantive macroeconomic conclusion independently of the dataset actually supplied.

12 Empirical Use: Template, Not Result

The repository includes an empirical workflow with local data loaders, catalog-based inputs, cached remote-source patterns, and multi-series configuration support. That means the codebase now supports a credible empirical template.

What it does not support yet is a strong article-level empirical claim on its own. The repository can process a real public dataset supplied locally or through a controlled cache workflow, then produce exceedance, recurrence, and clustering diagnostics. But whether those diagnostics justify a substantive empirical claim still depends on the actual dataset, threshold robustness, sample length, and interpretation discipline.

That distinction is important. The empirical layer is best described as ready for application, not as already delivering validated macroeconomic conclusions by itself.

13 Relationship to Standard Econometric Baselines

This project is not a rejection of standard econometrics. It is better understood as a complement. Baseline volatility-style and dependence summaries remain useful benchmarks. The repository explicitly keeps descriptive baseline comparisons because recurrence tools should be interpreted against familiar diagnostics rather than in isolation.

The incremental contribution is therefore specific: recurrence-oriented rare-event summaries provide information about tail occupation, clustering, and revisitation that standard baseline summaries may not foreground.

That complementarity is what `06_econometric_vs_recurrence_diagnostics.png` is for. The figure is strongest when used to compare descriptive lenses, not when used to claim that one framework fully dominates the other.

In a near-submission draft, this point should remain explicit: the repository compares diagnostic viewpoints, not full model classes in a horse race for universal superiority.

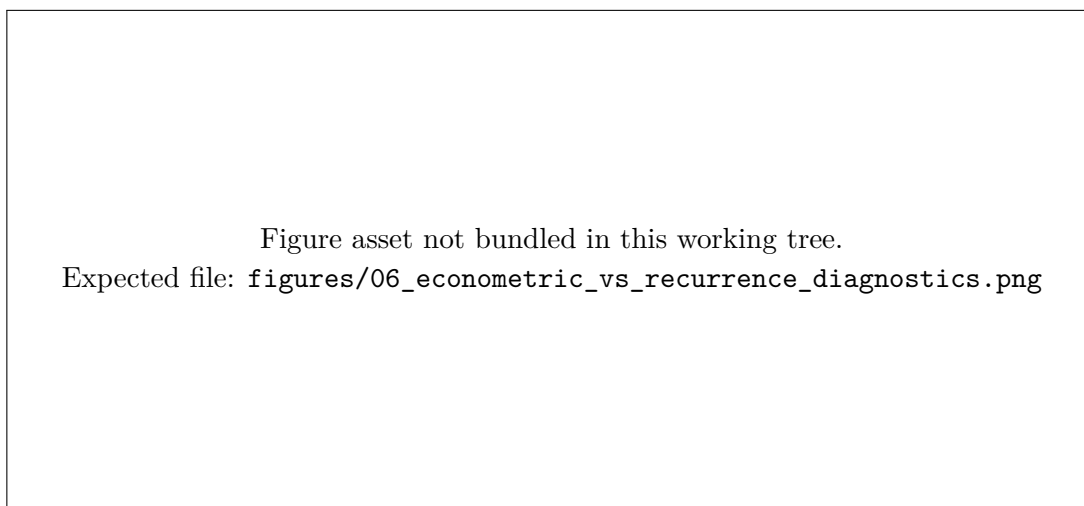


Figure 6: Baseline econometric versus recurrence-oriented diagnostics. This figure compares volatility-style baseline summaries with recurrence-oriented rare-event diagnostics. It supports the claim that the two views are complementary rather than interchangeable.

14 How the Figures Should Be Used in the Article

The article is strongest when each major claim is attached to one of the stable numbered figures and its source CSV:

- Figure 1 supports the computational workflow description.

- Figure 2 supports the central methodological motivation.
- Figure 3 supports descriptive recurrence timing statements.
- Figure 4 supports clustering comparisons.
- Figure 5 supports descriptive joint-stress timing.
- Figure 6 supports complementarity with standard baseline summaries.

Using the figures this way keeps the article disciplined. It prevents conceptual sections from drifting into claims that are not backed by saved repository outputs.

15 Limitations

The article should state the limitations plainly.

First, the repository does not establish deterministic laws for observed economies.

Second, it does not provide causal identification or policy attribution.

Third, synthetic benchmarks are methodological evidence, not empirical evidence.

Fourth, threshold choice matters materially, which is why threshold sensitivity belongs in the workflow rather than in a footnote.

Fifth, short samples and non-stationary real data can make rare-event summaries unstable, especially in the tail.

Those limits do not make the framework unhelpful. They define the zone in which it is honest.

16 Discussion

The practical contribution of the repository is therefore narrower than many crisis articles, but also more transparent. It does not promise a theory of everything, and it does not promise exact policy attribution. What it offers is a reproducible diagnostic language for thinking about how often stress returns, how it clusters, and whether it appears jointly across observables.

That narrower contribution is useful precisely because it fits the current state of the code. A computational research artifact is more persuasive when it makes a modest claim well than when it gestures toward a larger claim it cannot yet support.

17 Conclusion

The central contribution of the repository is not a grand theory of crises. It is a reproducible way to ask a sharper descriptive question: when extremes occur, do they behave like isolated events or like repeated returns to dangerous territory?

That question is worth asking because crisis damage is often driven by recurrence and clustering rather than by one large observation alone. The current codebase supports that claim through synthetic comparisons, recurrence diagnostics, multivariate stress summaries, and reproducible article-facing outputs with matching source files.

The right conclusion is therefore disciplined. Rare-event recurrence diagnostics expand the econometric description of danger. They do not replace standard econometrics, and they do not justify causal or deterministic claims. But they do provide a useful additional language for studying how stress returns, how it clusters, and how it appears jointly across observables.

In its current form, the repository is best understood as a reproducible methodological bridge: strong enough to support a careful article about recurrence diagnostics, transparent enough to show exactly where the evidence stops, and extensible enough to support future empirical applications built on real public datasets and tighter claim discipline.

References Placeholder

The final submission draft should include references in at least the following categories:

- foundational work on extreme value theory and recurrence in dynamical systems;
- applied references on extremal index estimation and declustering;
- standard econometric references for baseline volatility and persistence diagnostics;
- any data-source references used in a concrete empirical application;
- repository citation metadata for the computational artifact itself.

Until those references are inserted explicitly, the article should be treated as a polished draft rather than a final manuscript.